Pablo Morilla Cabello

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PabloMorillaCabello | Portfolio

Summary

Automation and robotics engineer with 2+ years of hands-on experience programming industrial robots and PLC systems for tier-1 automotive manufacturers. Proven track record delivering commissioning projects for Volvo Group, SCANIA, and Hitachi. Currently pursuing an MS in AI & Automation, combining deep industrial expertise with machine learning and computer vision skills.

Experience

Automation Engineer *Elektroautomatik* — *Skövde, Sweden*
April 2024 – June 2026

- Programmed and commissioned industrial robot cells and PLC-based automation lines for Volvo Group, SCANIA, and Hitachi.
- Developed reusable programming standards that reduced robot cell setup time across projects.
- Delivered on-site commissioning internationally, coordinating with cross-functional engineering teams.
- Specialized in RAPID programming, PLC logic design, and system integration for production environments.

Research Assistant *ASSAR Industrial Innovation Arena* — *Skövde, Sweden*
June 2023 – August 2023

- Built and integrated simulation software for a 3D industrial robotic system using modular architecture.
- Led a summer research project from planning through delivery, resulting in a peer-reviewed publication (DOI: 10.3233/ATDE240164).
- Explored modularization and interface standardization for production system simulation.

Projects

AGV Digital Twin *(Demo Video)* — *GitHub*

- Built a full-stack digital twin for an AGV with OPC UA integration, path planning, collision avoidance, and a web-based remote control interface.
- **Stack:** C, JavaScript, MongoDB, Node-RED, RobotStudio, CODESYS, OPC-UA.

ULTRA-ADDIN – RobotStudio Plugin *(Demo Video)* — *GitHub*

- Developed a C# plugin for ABB RobotStudio enabling modular, standardized robot cell configuration – reducing manual setup effort for simulation engineers.
- **Stack:** C#, RobotStudio SDK, Python.

Mobile Robot Navigation System *(Demo Video)* — *GitHub*

- Implemented autonomous path planning and collision avoidance on a PIERO mobile robot using sensor fusion and control algorithms.
- **Stack:** MATLAB, Simulink, Arduino.

Education

University West – MS in AI and Automation — *August 2025 – June 2026*

- Coursework in AI/ML, Deep Learning, Sensor Technology & Image Analysis, Big Data Processing, and Systems & Control.
- Hands-on training in automation systems, mechatronics, and robotics (ABB Robot Certificate).
- Work-Integrated Learning (WIL) Certificate – industry-integrated projects at Production Technology Centre (PTC).

University of Málaga – BEng in Robotics, Electronics, and Mechatronics *August 2019 – February 2024*

- EUR-ACE accredited (240 ECTS). Specialization in Robotics & Automation.
- Core training in control systems, industrial automation, PLC programming, analog/digital electronics, signal processing, and computer-based control.
- Robotics fundamentals, mechatronic design, theory of machines & mechanisms, power electronics, and networked systems architecture.
- Programming in C/C++, MATLAB, and embedded systems (Arduino, Xilinx FPGA). Computer vision and image processing.
- Mathematics, physics, statistics, and operations research foundations.

Högskolan i Skövde – BS in Industrial Engineering *August 2022 – June 2023*

- Industrial robotics and offline programming using ABB RobotStudio in real and virtual environments.
- Process planning, lean manufacturing principles, and product realization with a holistic engineering approach.
- Logistics simulation and industrial control engineering for production systems.

Skills

Programming: Python, C++, C, C#, JavaScript, RAPID, MATLAB, Arduino, LaTeX

Robotics & Automation: ABB RobotStudio, ROS2, PLC (TIA Portal, TwinCAT, CODESYS), OPC-UA, MQTT, Industrial Commissioning

AI & Vision: ML/DL Integration with Robotics, 2D/3D Computer Vision, OpenCV

Tools & Platforms: Git/GitHub, Docker, MongoDB, Node-RED, OpenCV, MATLAB/Simulink, AutoCAD

Languages: Spanish (Native), English (Fluent), Swedish (Basic – SFI)